Data_Foundations_Sales_Analysis

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```
[1]
import pandas as pd
import numpy as np

df = pd.read_csv("daily_product_sales_sample.csv")

print("CSV imported successfully!")
print(df)
```

```
CSV imported successfully!
         Date       Product  Units_Sold  Unit_Price      Store
0  2026-01-01      Notebook          25         120    Chennai
1  2026-01-02           Pen          40          15    Chennai
2  2026-01-03  Water Bottle          18         250  Bengaluru
3  2026-01-04      Notebook          30         120    Chennai
4  2026-01-05      Backpack           8         850  Hyderabad
5  2026-01-06           Pen          55          15  Bengaluru
6  2026-01-07  Water Bottle          22         250    Chennai
7  2026-01-08      Notebook          35         120  Hyderabad
8  2026-01-09      Backpack          12         850    Chennai
9  2026-01-10           Pen          48          15  Bengaluru
10 2026-01-11      Notebook          28         120    Chennai
```

```
[2]
sales_array = np.array(df["Units_Sold"])

print("Daily units sold array:")
print(sales_array)
```

```
Daily units sold array:
[25 40 18 30  8 55 22 35 12 48 28 20 10 52 32]
```

```
[3]
df["Daily_Sales"] = df["Units_Sold"] * df["Unit_Price"]

print(df[["Date", "Product", "Units_Sold", "Unit_Price", "Daily_Sales"]])
```

```
         Date       Product  Units_Sold  Unit_Price  Daily_Sales
0  2026-01-01      Notebook          25         120         3000
1  2026-01-02           Pen          40          15          600
2  2026-01-03  Water Bottle          18         250         4500
3  2026-01-04      Notebook          30         120         3600
4  2026-01-05      Backpack           8         850         6800
5  2026-01-06           Pen          55          15          825
6  2026-01-07  Water Bottle          22         250         5500
```

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```
[4]
print("Total units sold:", df["Units_Sold"].sum())
print("Minimum daily units sold:", df["Units_Sold"].min())
print("Maximum daily units sold:", df["Units_Sold"].max())
print("Average daily units sold:", df["Units_Sold"].mean())

print("Total revenue:", df["Daily_Sales"].sum())
print("Minimum daily revenue:", df["Daily_Sales"].min())
print("Maximum daily revenue:", df["Daily_Sales"].max())
print("Average daily revenue:", df["Daily_Sales"].mean())
```

```
Total units sold: 435
Minimum daily units sold: 8
Maximum daily units sold: 55
Average daily units sold: 29.0
Total revenue: 61425
Minimum daily revenue: 600
Maximum daily revenue: 10200
Average daily revenue: 4095.0
```

```
[5]
df.head()
```

	Date	Product	Units_Sold	Unit_Price	Store	Daily_Sales
0	2026-01-01	Notebook	25	120	Chennai	3000
1	2026-01-02	Pen	40	15	Chennai	600
2	2026-01-03	Water Bottle	18	250	Bengaluru	4500
3	2026-01-04	Notebook	30	120	Chennai	3600
4	2026-01-05	Backpack	8	850	Hyderabad	6800

```
df.tail()
```

	Date	Product	Units_Sold	Unit_Price	Store	Daily_Sales
10	2026-01-11	Notebook	28	120	Chennai	3360
11	2026-01-12	Water Bottle	20	250	Hyderabad	5000
12	2026-01-13	Backpack	10	850	Bengaluru	8500
13	2026-01-14	Pen	52	15	Chennai	780
14	2026-01-15	Notebook	32	120	Hyderabad	3840

```
print(df.shape)
```

```
(15, 6)
```

```
print(df.columns)
```

```
Index(['Date', 'Product', 'Units_Sold', 'Unit_Price', 'Store', 'Daily_Sales'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   Date         15 non-null     object
 1   Product      15 non-null     object
 2   Units_Sold   15 non-null     int64 
 3   Unit_Price   15 non-null     int64 
 4   Store        15 non-null     object
```

[10]
✓ 0s

```
df.describe()
```

	Units_Sold	Unit_Price	Daily_Sales
count	15.000000	15.000000	15.00000
mean	29.000000	264.000000	4095.00000
std	14.861263	314.240581	2857.28705
min	8.000000	15.000000	600.00000
25%	19.000000	67.500000	1912.50000
50%	28.000000	120.000000	3840.00000
75%	37.500000	250.000000	5250.00000
max	55.000000	850.000000	10200.00000

```
print(df.isnull().sum())
```

```
Date           0
Product        0
Units_Sold     0
Unit_Price     0
Store          0
Daily_Sales    0
dtype: int64
```

```
print("Number of duplicate rows:", df.duplicated().sum())
print(df[df.duplicated()])
```

```
Number of duplicate rows: 0
Empty DataFrame
Columns: [Date, Product, Units_Sold, Unit_Price, Store, Daily_Sales]
Index: []
```

```
df["Sales_Category"] = np.where(
    df["Daily_Sales"] >= 5000,
    "High",
    "Low"
)

print(df[["Date", "Product", "Daily_Sales", "Sales_Category"]]
```

```
           Date       Product  Daily_Sales Sales_Category
0    2026-01-01      Notebook         3000            Low
1    2026-01-02           Pen          600            Low
2    2026-01-03  Water Bottle         4500            Low
3    2026-01-04      Notebook         3600            Low
4    2026-01-05      Backpack         6800           High
5    2026-01-06           Pen          825            Low
6    2026-01-07  Water Bottle         5500           High
7    2026-01-08      Notebook         4200            Low
8    2026-01-09      Backpack        10200           High
9    2026-01-10           Pen          720            Low
10   2026-01-11      Notebook         3360            Low
11   2026-01-12  Water Bottle         5000           High
```

```python
print("Average daily revenue:",
      df["Daily_Sales"].mean())

print("Minimum daily revenue:",
      df["Daily_Sales"].min())

print("Maximum daily revenue:",
      df["Daily_Sales"].max())

print("\nNumber of days in each category:")
print(df["Sales_Category"].value_counts())

print("\nAverage revenue by category:")
print(df.groupby("Sales_Category")["Daily_Sales"].mean())
```

```
Average daily revenue: 4095.0
Minimum daily revenue: 600
Maximum daily revenue: 10200

Number of days in each category:
```

```
Average daily revenue: 4095.0
Minimum daily revenue: 600
Maximum daily revenue: 10200

Number of days in each category:
Sales_Category
Low     10
High     5
Name: count, dtype: int64

Average revenue by category:
Sales_Category
High    7200.0
Low     2542.5
Name: Daily_Sales, dtype: float64
```

```
high_quantity = df[df["Units_Sold"] > 30]

print(high_quantity)
```

```
           Date    Product  Units_Sold  Unit_Price      Store  Daily_Sales  \
1    2026-01-02        Pen          40          15    Chennai          600
5    2026-01-06        Pen          55          15  Bengaluru          825
7    2026-01-08   Notebook          35         120  Hyderabad         4200
9    2026-01-10        Pen          48          15  Bengaluru          720
13   2026-01-14        Pen          52          15    Chennai          780
14   2026-01-15   Notebook          32         120  Hyderabad         3840

    Sales_Category
1              Low
5              Low
7              Low
9              Low
13             Low
14             Low
```

```
high_revenue = df[df["Daily_Sales"] > 5000]

print(high_revenue)
```

```
           Date       Product  Units_Sold  Unit_Price      Store  Daily_Sales  \
4    2026-01-05      Backpack           8         850  Hyderabad         6800
6    2026-01-07  Water Bottle          22         250    Chennai         5500
8    2026-01-09      Backpack          12         850    Chennai        10200
12   2026-01-13      Backpack          10         850  Bengaluru         8500

    Sales_Category
4             High
6             High
8             High
12            High
```

```
notebook_sales = df[df["Product"] == "Notebook"]

print(notebook_sales)
```

```
           Date   Product  Units_Sold  Unit_Price      Store  Daily_Sales  \
0    2026-01-01  Notebook          25         120    Chennai         3000
3    2026-01-04  Notebook          30         120    Chennai         3600
7    2026-01-08  Notebook          35         120  Hyderabad         4200
10   2026-01-11  Notebook          28         120    Chennai         3360
14   2026-01-15  Notebook          32         120  Hyderabad         3840

    Sales_Category
0              Low
3              Low
7              Low
10             Low
14             Low
```

```
filtered_data = df[
    (df["Units_Sold"] > 20) &
    (df["Store"] == "Chennai")
]

print(filtered_data)
```

```
         Date       Product  Units_Sold  Unit_Price    Store  Daily_Sales  \
0  2026-01-01      Notebook          25         120  Chennai         3000
1  2026-01-02           Pen          40          15  Chennai          600
3  2026-01-04      Notebook          30         120  Chennai         3600
6  2026-01-07  Water Bottle          22         250  Chennai         5500
10 2026-01-11      Notebook          28         120  Chennai         3360
13 2026-01-14           Pen          52          15  Chennai          780

   Sales_Category
0             Low
1             Low
3             Low
6            High
10            Low
```

```
df.to_csv("analyzed_product_sales.csv", index=False)

print("Analyzed CSV saved successfully!")
```

```
Analyzed CSV saved successfully!
```

```
from google.colab import files

files.download("analyzed_product_sales.csv")
```